

Chapter 1

Explainability in Reinforcement Learning

Today, the majority of artificial intelligence agents with which humans interact are deployed in relatively controlled and low-risk environments. These agents are currently mainly used in simulated, digital, or non-critical impact settings, such as games, recommendation systems, online advertising, or computer system optimization. For example, agents capable of playing games such as Go operate in settings where potential errors remain limited in terms of consequences. Even more autonomous systems, such as agents based on large language models, remain strongly constrained by human supervision mechanisms and are often restricted to tasks such as code generation or information retrieval. Safety-critical applications integrating machine learning components are still relatively rare.

However, many potential high-impact applications such as drone control, autonomous delivery, autonomous vehicle driving, adaptive urban traffic management, optimization of smart power grids, or control of industrial robotic systems in real-world environments would require significantly higher levels of autonomy. Introducing autonomous agents in such environments, requires the establishment of strong guarantees. Without satisfying several essential requirements, deploying autonomous agents in the real world remains hardly feasible. These requirements include in particular:

- validating the system’s functioning before deployment,
- identifying the origin of a malfunction in case of failure,
- and certifying that an observed failure will not occur again.

The main challenge lies in the fact that the performing agents currently available rely on both reinforcement learning and deep learning. As a result, these agents are unable to provide explanations of the mechanisms underlying their decisions. This lack of explainability makes their use difficult in safety-critical contexts. To address this limitation, a research field has emerged: Explainable Reinforcement Learning (XRL), a subfield of Explainable Artificial Intelligence (XAI).

This chapter aims to present the main existing XRL techniques. To this end, we first define what is meant by an “explanation” in the context of XAI. We

then propose a taxonomy for organizing the XRL research field. Finally, we review the state of the art following this organization.

1.1 Explanation in Artificial Intelligence

In this section, we define what an explanation is in the context of XAI and how it can be obtained. We first introduce the concept of explanation in a general setting. We then describe how this concept applies to XAI. Finally, we present explainability methods and the properties that characterize an explanation in the context of XAI.

1.1.1 Explanation

Establishing what is meant by an explanation is challenging due to the fundamentally multidisciplinary nature of the concept. When we refer to explanation, we are inherently referring to an explanation provided to a human. As soon as humans are involved, the problem is no longer purely computational or mathematical. It involves several fields of research, such as philosophy, psychology, and sociology. So far, these disciplines have not converged on a unified definition of explanation that can be directly applied to XAI. For this reason, we introduce here a working definition of explanation.

An explanation involves two entities: the object being explained and the receiver of the explanation. The receiver of the explanation is a human, who will be referred to as the observer throughout this manuscript. The observer aims to understand how the object works and, in doing so, builds a mental model of its functioning. In this context, an explanation is what allows the observer to refine and correct their mental model of how the system operates.

Definition 1.1.1: Object

The object of an explanation is the system is to be understood by an observer.

Definition 1.1.2: Observer

The observer is the human who aims to understand a object. The observer possesses a mental model of the object being explained. This mental model is an internal representation of how the observer believes the system operates. This representation may be incomplete or inaccurate.

Definition 1.1.3: Explanation

An explanation is any information that enables an observer to refine or correct their mental model of the object of the explanation.

This definition of explanation is intentionally broad, allowing it to encompass the different types of explainability methods considered so far. We adopt this minimal definition to provide a common framework for evaluating the effectiveness of explanations. Explanations do not all provide the same level of value: describing a marginal aspect of a system is fundamentally different from correcting an observer's misunderstanding of a key mechanism.

1.1.2 Explainable Artificial Intelligence

In the context of XAI, the object of explanation is an artificial intelligence system. More precisely, XAI emerged as a response to the success of deep learning models. These models achieve high performance through connectionist architectures based on deep neural networks. In these architectures, information is represented in a distributed manner across the network through patterns of activation and connections between computational units. Unlike symbolic approaches, individual components of neural networks cannot be directly associated with human-defined concepts. The knowledge encoded by these models cannot be understood simply by inspecting their structure.

From this starting point, two types of approaches emerge:

- **Substitutive explainability:** The idea is that there is nothing to explain in a trained neural network. There is no semantic meaning associated with each computational component, and therefore nothing to interpret. This corresponds to the so-called "black box" view of neural networks. Under this assumption, XAI consists of replacing as much of the deep learning system as possible with alternative AI methods.
- **Intrinsic explainability:** It is argued that neural networks can be explained. The limitation is not considered to be a lack of interpretability, but rather the absence of tools to analyze their functioning. According to this perspective, neural networks contain structures and mechanisms that can be studied in order to understand how they operate.

This manuscript follows the intrinsic explainability perspective. From this point of view, we define the meaning of the term explainable in the context of XAI. Taken literally, explainable refers to something that can be explained. However, according to the definition of explanation introduced previously, any system can be explained 1.1.3. Therefore, from our perspective, describing an AI system as either explainable or not explainable does not provide a meaningful distinction. For consistency with the existing literature, we continue to use the term XAI to refer to this research field.

Having defined the term explainable, we now define the closely related notion of interpretable. In our view, interpretability is a property of information. Information is considered interpretable when it can be directly understood by a human without requiring any additional transformation process. Consequently, an interpretable AI system refers to a system whose functioning can be understood through direct analysis of its internal mechanisms.

1.1.3 Explainability Method

Now that the general framework of XAI has been established, we need to further detail the properties of an explanation in this context. An explanation in the field of XAI can be characterized by three components: the source, the form, and the subject. These three aspects are determined by the explainability method used to generate the explanation.

The characteristics of an explanation can be described through three aspects:

- **Source:** refers to the origin of the information used by the explainability method. By itself, this information is not necessarily interpretable by hu-

mans. The explainability method transforms this raw source of information into an explanation that can be understood by an observer.

- **Form:** refers to the representation through which the explanation is communicated. An explanation can take different forms, such as text, images, computer code, mathematical equations, or statistical. The form defines how the information is presented to the observer and must allow human interpretation.
- **Subject:** refers to the component of the AI system that is being explained. In XAI, this distinction is commonly associated with the difference between local and global explanations. A local explanation describes why an AI system produced a specific output in a given situation. A global explanation aims to describe the general functioning of the AI system.

An explainability method can therefore be viewed as a transformation process that maps a source of information into an explanation characterized by a specific form and subject. Current research in XAI focuses on developing new methods capable of performing this transformation in order to overcome the lack of interpretability associated with deep learning models.

1.2 Taxonomy of XRL Methods

Now that the concept of XAI has been defined, we can focus on its application to reinforcement learning. In this section, we propose an organization of XRL methods based on the XAI perspective introduced previously, while taking into account the specificities of RL. For this purpose, we rely on a review of the XRL literature, which allows us to identify the different sources, forms, and subjects that explanations can take in reinforcement learning.

1.2.1 Subject

The first question to address is the subject of explanation in the context of reinforcement learning. To answer this question, it is necessary to identify the different components involved in a reinforcement learning system and determine which of them constitutes the object of explanation. A reinforcement learning system can be described through two main components: the environment, formalized as a Markov Decision Process (MDP), and the agent's policy.

The environment defines the dynamics in which the agent evolves, including the possible states, actions, and transitions. Environments are simulations designed by humans, and their dynamics are explicitly defined. Therefore, it is assumed that the structure of the environment, as well as the agent's perception and action capabilities, are known to the observer and do not require explanation. These components are designed by humans and are therefore considered understandable by default.

The main object of interest is therefore the agent's policy, denoted π . The policy represents the decision-making mechanism of the agent. It is a function that maps observations $\mathcal{O}$ to actions $\mathcal{A}$ and is learned through interaction with the environment in order to maximize the expected cumulative reward. The policy is the component in which the learning process is encoded and therefore corresponds to the artificial intelligence system that XRL aims to explain.

Explaining a policy introduces additional challenges compared with classical XAI settings. In traditional XAI problems, the objective is to explain an isolated decision, such as determining which features of an image led to a classification result. In contrast, reinforcement learning introduces a temporal dimension into the decision-making process. An action is not only determined by the current observation but also influences future states and rewards. Therefore, understanding the behavior of an agent may require considering a sequence of interactions between the agent and its environment.

This temporal dimension leads to a distinction between two objectives of explanation in XRL:

- **Decision explanations:** focus on identifying the elements of the current observation that influenced the action selected by the policy. This type of explanation is close to classical XAI approaches, as it analyzes the relationship between inputs and outputs while abstracting away the sequential nature of the decision process.
- **Strategy explanations:** aim to characterize the agent’s behavior over time. They focus on sequences of decisions and interactions with the environment in order to understand how the agent achieves its long-term objective. This type of explanation is specific to reinforcement learning, as it addresses the temporal and goal-oriented nature of the learning process.

1.2.2 Source

We identify three sources of explanation in reinforcement learning: policies, trajectories, and value functions. These sources are obtained during or after the learning process and contain information about the agent.

- **Policies:** policies are functions that generate a distribution over the action space for a given observation. The objective of reinforcement learning is to learn a policy that maximizes the expected cumulative future rewards. During the learning phase, the policy is iteratively improved through interactions with the environment.
- **Trajectories:** trajectories represent sequences of successive interactions between the agent and the environment. At each step, the agent selects an action based on its observation, which modifies the environment and produces a reward as well as a new observation. This process continues until the end of the trajectory.
- **Value functions:** value functions provide predictions about the future outcome of a trajectory from a given observation. The most commonly used value function is the critic function, which estimates the expected sum of future rewards. This prediction guides the learning process, as the agent updates its policy to maximize the estimated future return. However, other types of value functions can also be used as sources of explainability.

1.2.3 Form

The form of an explanation is the aspect that varies the most between XRL methods. Based on our review of the literature, we identify six forms of explanation, although other forms may potentially exist:

- 199 • **Policy Model:** an interpretable model of the agent’s policy function.
- 200 • **Interaction Model:** an interpretable model of the interactions between
- 201 the agent and the environment.
- 202 • **Agent Trajectory:** a representation of a sequence of successive interac-
- 203 tions between the agent and the environment.
- 204 • **Trajectory Prediction:** a prediction about the future evolution of the
- 205 trajectory produced by a value function.
- 206 • **Counterfactual:** a representation of a modified observation obtained by
- 207 perturbing the original observation in order to produce a different action
- 208 from the agent.
- 209 • **Saliency Map:** a weighting of the input features representing their
- 210 influence on the output of the policy function.

211 Among the different components characterizing an explanation, the form is
 212 the element that differs most significantly from one explainability technique to
 213 another. Therefore, the following section is organized according to the form
 214 of explanations and presents the existing explainability methods based on this
 215 criterion.